Supplementary Methods and Prompts

Title: Bilingual Retrieval-Augmented Access to ANVISA Regulatory Documents

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Purpose

This file provides reproducibility details that are too technical for the main manuscript. It documents corpus processing, article-aware chunking, retrieval, reranking, bilingual query support, generation prompts, refusal rules, evaluation metrics, and automated judging. Source files used to verify these details are included in evidence_artifacts/code/.

Software Files Used for Reproducibility

File	**Role**
evidence_artifacts/code/config.py	Models, paths, chunking constants, retrieval constants, reranker constants
evidence_artifacts/code/prompts.py	Final answer-generation prompt
evidence_artifacts/code/retrieval.py	Hybrid retrieval, BM25 tokenization, rank fusion, cross-encoder reranking, anchors
evidence_artifacts/code/publication_eval.py	Evaluation loop, generation loop, metrics exported to CSV, run manifests
evidence_artifacts/code/answer_quality.py	Citation extraction, citation validation, AI-judge prompt
evidence_artifacts/code/ evaluation_statistics.py	Article retrieval metrics and bootstrap confidence intervals

Corpus Processing

The source corpus consisted of six selected ANVISA regulatory documents. The files were processed into cleaned plain-text files before indexing. Cleaning removed HTML/MHTML markup residue, page boilerplate, and formatting artifacts while preserving article labels and regulatory structure. The final cleaned corpus was the only source used for retrieval and answer generation.

The closed-corpus design was intentional. The system did not search live ANVISA pages, submission portals, current form repositories, fee tables, private company records, or documents

outside the local index. A question asking for information outside the indexed documents was treated as unanswerable, even when it was regulatory in topic.

Source verification identified one metadata issue: the local file name RDC_406_2020_Estudos_de_Estabilidade.txt suggested stability studies, but the document content was RDC 406/2020 on pharmacovigilance. All manuscript and supplement tables therefore describe that document by official regulation number and topic.

Article-Aware Chunking

The final index used article-aware chunking rather than arbitrary fixed-size chunks. The target unit was one regulatory article per chunk when possible. Nearby chapter, section, document, and citation metadata were retained with each chunk.

Chunking constants:

Setting	**Value**
Single-article chunk maximum	500 tokens
Long-article chunk size	400 tokens
Long-article overlap	60 tokens
Embedding maximum length	512 tokens

Final index audit:

Measure	**Value**
Source documents	6
Detected articles	382
Total chunks	481
Single-chunk articles	359
Long articles split into multiple chunks	23
Split chunks from long articles	150
Median chunk length	155 tokens
Maximum chunk length	503 tokens
Chunks above 700 tokens	0

Embeddings and Vector Index

The embedding model was sentence-transformers/all-MiniLM-L6-v2. In the final package, the model path was recorded from the local cache:

models--sentence-transformers--all-MiniLM-L6-v2/snapshots/
c9745ed1d9f207416be6d2e6f8de32d1f16199bf

The vector store was rebuilt after corpus cleaning and article-aware chunking. The pipeline manifest recorded 382 articles, 481 chunks, and six sources.

Hybrid Retrieval

The final retriever combined dense vector retrieval and keyword retrieval.

Retrieval settings:

Setting	**Value**
Dense vector candidate depth	100
BM25 candidate depth	100
Rank-fusion candidate depth	100
Reciprocal-rank-fusion constant	60
Final reranked passages sent to answer model	5
Reranker batch size	8

BM25 used normalized lexical tokens. Text was normalized by removing accents, lowercasing, and extracting alphanumeric tokens. BM25 was applied to searchable text that combined chunk text and metadata fields, including regulation identifier, document aliases, document title, citation label, chapter, section, and topics.

Dense and BM25 candidates were merged with reciprocal rank fusion:

score = sum(1 / (60 + rank))

Candidates were sorted by fused score and then reranked.

Reranking

The reranker was cross-encoder/mmarco-mMiniLMv2-L12-H384-v1. In the final local cache, the model path was:

models--cross-encoder--mmarco-mMiniLMv2-L12-H384-v1/snapshots/
1427fd652930e4ba29e8149678df786c240d8825

The reranker scored pairs of query text and searchable node text. The top five reranked passages were used to build the answer context.

Additional deterministic anchors were used for specific regulatory patterns:

- Source-identification questions could force an Article 1/source-scope anchor when the question asked which document covered a topic.
- Named regulation scope questions could anchor the corresponding regulation's Article 1.
- Certain exact legal patterns added a targeted anchor when previous development runs showed that the relevant article could be missed by generic reranking.
- Neighbor articles were added for selected scope, definition, exemption, responsibility, and medical-device questions when adjacent articles were likely to be needed for completeness.

These anchors were rule-based retrieval aids. They did not add facts to the answer. The answer model still had to use retrieved context and article citations.

English Query Support

English questions were answered in English, but retrieval was allowed to include Portuguese anchors because the source documents were in Portuguese. The original question remained visible to the answering model. Conservative terminology anchors were used for regulatory terms:

English term	**Portuguese retrieval anchor**
clinical trial	ensaio clinico
sponsor	patrocinador
biological product	produto biologico
medical device	dispositivo medico
stability study	estudo de estabilidade
pharmacovigilance	farmacovigilancia

The purpose was retrieval support, not translation of the benchmark answer.

Answer Model

The answer model was models/gemini-flash-lite-latest. Generation used the retrieved context and the original question. The answer had to be in the same language as the question. Answers were post-processed only to normalize article citation brackets when the answer added paragraph or item suffixes to a retrieved article-level citation.

Final Answer Prompt

The final answer prompt was:

```text
You are an ANVISA regulatory evidence assistant. Answer the question using only the retrieved context below.

Rules:

1. Do not use outside knowledge or infer facts not stated in the context.

2. Answer in the same language as the question.

The retrieved context may be Portuguese; still answer in English when the question is English.

If the question is written in Portuguese, answer in Portuguese. Do not translate Portuguese questions into English.

3. Cite every material statement inline using the context's citation_label exactly, formatted in square brackets, for example [RDC 945/2024, Art. 23].

Put each citation in its own pair of brackets.

Do not add paragraph, item, section, page, or line suffixes unless they are part of the retrieved citation_label itself.

Every non-refusal answer must include at least one citation containing "Art."; do not rely only on a preamble or document-level citation when article evidence is available.

4. Never invent a document number, article, section, deadline, or requirement.

5. If the context does not contain enough evidence to answer the exact question, respond only with the appropriate sentence below and do not add a citation:

Portuguese: Os documentos recuperados nao fornecem informacao suficiente para responder.

English: The retrieved documents do not provide enough information to answer.

Treat related-but-incomplete context as insufficient. Do not substitute a nearby general rule, procedure, system name, deadline, definition, or article when the
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question asks for a more specific fact that is absent.
This includes questions asking for current forms, form models, exact URLs, current fees or values, private contract terms, company-specific submission status, personal contact information, market prices, reimbursement, taxes, patents, future approvals, or advice that is not explicitly stated in the retrieved regulatory text.
6. Before answering, silently classify the retrieved evidence as one of:
sufficient: it directly answers every material part of the question;
partial_related: it discusses the topic but omits the requested specific fact;
insufficient: it does not contain relevant evidence.
Answer only when evidence is sufficient. For partial_related or insufficient, return only the refusal sentence from rule 5.
7. For yes/no or false-premise questions, answer with a cited negative only when the retrieved context directly contradicts the premise. Use "Nao" for Portuguese questions and "No" for English questions. If the context merely lacks the premise or does not address it, use the refusal sentence instead.
8. Prefer a concise direct answer, but answer every material part of the question. Include deadlines, formats, lists of required documents, conditions, exceptions, and consequences when the question asks for them and the context provides them.
When the relevant article contains numbered items, lettered subitems, or paragraphs such as I, II, III, a), b), §1, §2, read the whole retrieved article before answering. For "what documents", "what information", "which are excluded", "which are exempt", or similar list questions, include every relevant category from the retrieved article. You may group related items to stay concise, but do not omit early or later items from the same article.
9. When the question asks which regulatory document covers a topic, matching article

text plus its regulation_id or citation_label is sufficient evidence to name that document. Cite the most directly relevant retrieved article.

10. For purpose or scope questions, supplement preamble evidence with relevant objective, scope, or definitions articles when they are retrieved.

Final validation: unless the response is the refusal sentence or answers a publication date found only in the preamble, it is invalid if it does not contain the literal text ", Art." inside at least one square-bracket citation. Revise it using retrieved article evidence before returning it. Never invent an article citation for preamble-only facts.

Retrieved context:

{context_str}

Question: {query_str}

Answer:

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Refusal Rule

The refusal sentence was language-specific:

Question language	**Refusal sentence**
Portuguese	Os documentos recuperados nao fornecem informacao suficiente para responder.
English	The retrieved documents do not provide enough information to answer.

Refusal was required when context was absent, irrelevant, or only partially related. The final prompt explicitly classified related-but-incomplete evidence as insufficient. This was added because the first AI-authored prospective stress test showed that the system sometimes answered current operational questions from nearby general regulatory text.

Benchmark Design

The 200-question benchmark contained 100 bilingual pairs and 200 rows:

Bucket	Count
Portuguese answerable	70
English answerable	70
Multi-article or exception	20
Unanswerable or adversarial	40

Language counts were balanced: 100 Portuguese rows and 100 English rows. The benchmark audit recorded 78 unique gold article citations. Portuguese and English members of each pair shared the same data split to avoid leakage across languages.

Split counts:

Split	Rows
Held-out test	75
Calibration	20
Development remainder	5

The AI-authored prospective stress test contained 50 questions, organized as 25 bilingual pairs. The refusal-focused AI-authored test contained 30 questions, organized as 15 bilingual pairs.

Evaluation Outputs

For each row, the evaluation script saved:

- question id
- pair id
- language
- category
- answerability
- gold source documents
- gold articles
- gold citations
- raw vector sources and citations
- BM25 sources and citations
- fused sources and citations
- final retrieved sources and citations

- retrieval query and rerank query
- rewrite status
- answer text
- answer citations
- refusal detection
- citation validity
- answer term coverage
- AI-judge scores when judging was enabled

Manual-review templates were also exported with fields for human source correctness, groundedness, completeness, citation correctness, safety/refusal correctness, error type, and comments. These were retained as audit scaffolds even though the final evaluation used AI judging only.

Retrieval Metrics

Source-level metrics:

- source hit: whether any gold source document was retrieved in the evaluated top-k
- source MRR: reciprocal rank of the first gold source document

Article-level metrics:

- article Recall@k: fraction of gold article citations retrieved in top-k
- article Precision@k: fraction of top-k retrieved citations that were gold citations
- article MRR@k: reciprocal rank of first gold citation
- article nDCG@k: ranking quality for gold citation placement

Gold and retrieved labels were normalized by lowercasing, removing accents, and collapsing whitespace before comparison.

Answer Metrics

Answer-level metrics:

- citation presence: whether the answer contained a recognized bracketed citation
- citation retrieved validity: fraction of citation-like brackets matching retrieved citation labels
- article citation presence: whether at least one answer citation contained an article label
- answer term coverage: fraction of expected answer terms found in the generated answer
- refusal detected: whether the answer matched the defined refusal patterns

Automated AI Judge

The AI judge used models/gemini-flash-lite-latest with temperature 0.0. The judge received the question, system answer, expected answer, answerability flag, and retrieved evidence. It returned JSON with four numeric scores from 0 to 1:

- faithfulness
- citation_correctness
- completeness
- refusal_correctness

The judge prompt was:

```text

Evaluate a regulatory RAG answer. Return only JSON with numeric scores from 0 to 1

and a short rationale.

Scoring:

- faithfulness: every factual claim is supported by retrieved evidence.
- citation_correctness: citations exist when an answer is given, refer to retrieved

evidence, and support the associated claim.

The system uses article-level citation labels only. If the supporting sentence

is in a paragraph, item, or subitem inside a retrieved cited article, the

article-level citation is correct; do not penalize the answer for omitting

paragraph, item, section, page, or line suffixes.

- completeness: the answer covers the expected answer's essential information,

including material conditions or exceptions.

- refusal_correctness: score 1 when an unanswerable question is refused without an

unsupported answer, or when an answerable question is answered rather than refused.

Question answerable from corpus: {answerable}

Question: {question}

Expected answer: {expected_answer}

System answer: {answer}

Retrieved evidence:
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{evidence}

JSON fields: faithfulness, citation_correctness, completeness,

refusal_correctness, rationale

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The AI judge was used as a structured technical scoring tool. It was not treated as an independent regulatory expert panel. The answer model and automated judge used the same Gemini model family, so same-model-family grading may overestimate answer quality.

Confidence Intervals

When confidence intervals were reported, the method was pair-clustered percentile bootstrap:

Setting	**Value**
Confidence level	95%
Resamples	2000
Seed	20260619
Cluster key	bilingual pair id

Pair clustering was used because Portuguese and English questions were written as paired items.

Reproducibility Boundary

The package supports reproduction of the reported technical evaluation if the same code, corpus files, model identifiers, benchmark files, API model behavior, and package versions are available. A conda environment export and pip freeze file are included as environment.lock.yml and requirements-freeze.txt. Exact generated answers may still vary because hosted LLM behavior can change over time even when the model name and prompt are fixed. The archived per-question CSVs are therefore part of the evidence record and should be used to audit the reported results.

Operational Failure Thresholds

The primary reported metrics were continuous means. They were not converted into a single binary success claim for the manuscript. The evaluation code nevertheless used operational flags to identify rows requiring review.

Answerable rows were flagged when the retrieved gold source was missing, article retrieval was incomplete, retrieved expected-term coverage was below 0.50, generated answer-term coverage was below 0.50, citations were absent, cited labels were not among retrieved labels, or the system refused an answerable question.

AI-judge faithfulness below 0.80 was flagged as unfaithful. The English completeness diagnostic experiment used AI-judge completeness below 0.80 to identify low-completeness answerable English rows. Unanswerable rows were flagged when the literal refusal sentence was not detected.

These thresholds were used for technical error analysis and targeted repair. They were not regulatory acceptance criteria.